

Which of the following regulatory mechanisms helps increase net glucose catabolism in the liver after a meal?

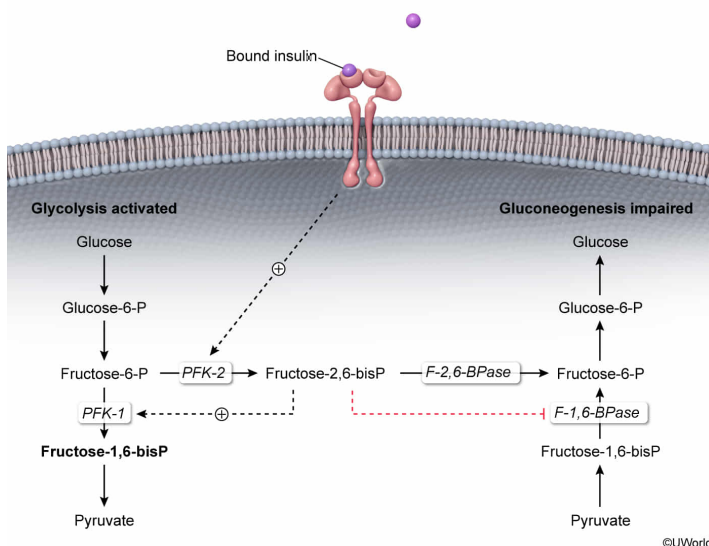
- ☐ A. Inhibition of hexokinase by glucose-6-phosphate [10%]
- ☐ B. Allosteric suppression of phosphofructokinase-1 by ATP binding [10%]
- ☐ C. Hormonal suppression of fructose-2,6-bisphosphate synthesis [23%]
- ✓ ☒ D. Allosteric inhibition of fructose-1,6-bisphosphatase activity [54%]

Correct

55% answered correctly

Explanation:

Regulation of glycolysis and gluconeogenesis by insulin in the fed state



Glucose uptake into cells and its subsequent metabolism are controlled primarily by the hormones **insulin and glucagon**. The binding of insulin and glucagon at the cell membrane stimulates or suppresses the biochemical pathways involved in the uptake, breakdown, and storage of glucose. Insulin stimulates **glycolysis**, which consists of a series of reactions that break down glucose into pyruvate to produce ATP. Glucagon acts in the reverse by activating **gluconeogenesis** in the **liver**, which results in the synthesis of glucose from pyruvate and other precursor molecules.

The **well-fed state** is characterized by the **release of insulin** in response to **high levels of glucose** in the blood. Insulin stimulates glucose catabolism (glycolysis) by activating the synthesis of fructose-2,6-bisphosphate (F2,6BP). F2,6BP allosterically activates phosphofructokinase-1 (PFK-1), which catalyzes the rate-limiting step of glycolysis. In the liver, **F2,6BP** also inhibits gluconeogenesis by **allosterically inhibiting** the catalytic activity of the gluconeogenesis enzyme **fructose-1,6-bisphosphatase**. By **inhibiting gluconeogenesis** in the liver, the glycolysis-gluconeogenesis balance shifts, leading to *increased* glycolysis and therefore **increased net glucose catabolism**.

(Choice A) **Hexokinase** converts glucose into glucose-6-phosphate in the first step of glycolysis. Inhibition of hexokinase would *reduce* glucose catabolism.

(Choice B) Phosphofructokinase-1 catalyzes the conversion of fructose 6-phosphate to fructose-1,6-bisphosphate in glycolysis. Inhibiting this process would slow down glucose catabolism.

(Choice C) Activation, not suppression, of F2,6BP synthesis increases glucose catabolism.

Educational objective:

Insulin stimulates glycolysis in the liver by activating the enzyme phosphofructokinase-2, which synthesizes fructose-2,6-bisphosphate. Fructose-2,6-bisphosphate allosterically activates phosphofructokinase-1 and inhibits fructose-1,6-bisphosphatase, which stimulates glycolysis and reduces gluconeogenesis.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions